SCI-MAP

TolTEC Ordinary Mapmaking
and Observation Coaddition
Scientific Basis, Response, and Validity

Scientific contract version: v0.1
Science-team rationale revision: r0.4
Status: integrated scientific authority draft
2026-08-26

This document explains the map estimator, its response and uncertainty, the meaning of support and validity, and the scope of observation coaddition for TolTEC scientists. The separate formal contract remains the normative authority for the 52 requirements, 25 predictions, exact states, provenance, and conformance procedure.

Scientific owner: Grant Wilson

Executive summary and current status

For a map pixel p, SCI-MAP forms the coefficient-weighted average

$$\widehat{x}_p = \frac{\sum_i G_{pi}\omega_i x_i}{\sum_i G_{pi}\omega_i} = \frac{N_p}{Q_p}, \tag{1}$$

using only realized PTC samples that are available under PTC's declared identity and scope, have valid ALIGN/AST coordinates, and pass MAP's admission conditions. Here G_{pi} describes how sample i is assigned or divided among pixels, and ω_i is the PTC-supplied MAP-facing analysis coefficient. Pixels without adequate valid support are unavailable; they are not measurements of zero sky brightness.

For compatible observation maps on one common grid, the v0.1 coadd is

$$\widehat{s}_p = \frac{\sum_o u_{op}\widehat{x}_{op}}{\sum_o u_{op}}, \tag{2}$$

where every admitted observation coefficient u_{op} is finite and positive. An observation bundle is admitted atomically: its signal, coordinates, response, uncertainty, validity, and provenance must be mutually compatible before any of its valid pixels contribute. Pixel-level support then determines where the admitted observation contributes.

Both equations are linear only *conditional on fixed* sample and detector membership, projection, coefficients, support policy, response state, geometry/WCS, calibration, and upstream processing. If any of those states is estimated from the same data, unconditional bias or covariance requires a joint model or resampling result.

Intended scientific estimator	Positive-coefficient ordinary map and compatible common-grid observation coadd, Equations (1)–(2).
Upstream input status	MAP always consumes a realized PTC product; there is no CAL-to-MAP fallback. ALIGN/AST owns coordinate facts, PTC owns product availability and producer-local validity, MAP owns MAP admission, and VAL may evaluate named rules without authoring them.
Implementation conformance	Not assessed in this authorship revision.
Response/representation validation	Not established; response injections and WCS round trips remain required evidence.
Observational validation	Not established; amplitude, shape, position, covariance, and repeatability require astronomical controls.
Production readiness	Not claimed and cannot be inferred from the algebraic contract.

Contents

1 What SCI-MAP estimates

SCI-MAP estimates a map-domain average of the *realized PTC-transformed* sample quantity. It does not estimate unconvolved sky truth. The physical meaning of the result is therefore inherited from the complete upstream CAL-to-PTC chain: what the ordinary `xs` sample represents, how it was calibrated and transformed, what sky coordinate it has, and whether the required PTC product exists under its declared producer-local validity.

The numerical signal scale may be the declared point-source-equivalent mJy beam^{-1} convention inherited from SCI-CAL. MAP does not create absolute calibration, choose a beam convention, or complete an unresolved broadband photometric definition. A numerical map may consequently exist while literal point-source-peak interpretation remains unavailable because the response for the declared processing chain is incomplete. The same caution applies to extended-source fidelity, integrated flux, and absolute offsets.

The map response is the full upstream sky-to-sample response followed by the map operator. Writing the selected processed-sample vector as $\boldsymbol{y} = (x_i)$, the map and its response are

$$\widehat{\boldsymbol{x}} = A\boldsymbol{y}, \qquad R = AH, \tag{3}$$

where H maps a declared sky model into processed samples and A maps those samples into valid pixels. Preserving a constant *sample vector* does not prove preservation of a constant sky mode: an upstream baseline, common-mode removal, or filter may already have removed it.

A stored kernel is a realized response only when its template, sample-domain parent, and processing lineage prove the relation in Equation (3). Otherwise it is a tracer or has an explicit limited or unavailable status. That state does not invalidate the numerical map or prohibit later analysis; it only limits the claims made by the original bundle. A later simulated response or response-corrected map is a new versioned product bound to the original MAP product and processing identity.

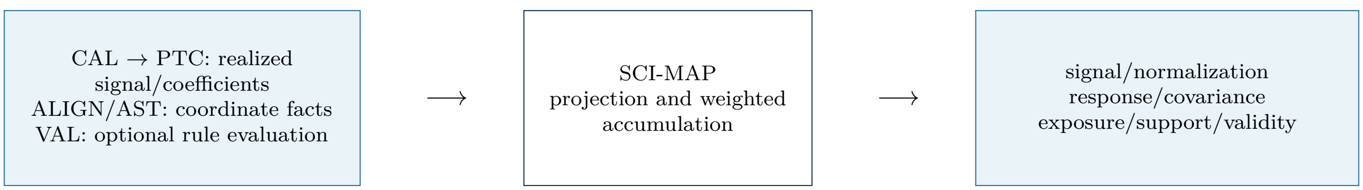

Figure 1: SCI-MAP transforms declared upstream quantities; it does not repair or reinterpret them.

2 From samples to pixels

For each candidate sample i, the projection coefficient G_{pi} is the assignment or fraction by which that sample contributes to pixel p. One-hot projection has one nonzero $G_{pi} = 1$. Fractional projection may divide a sample among neighboring pixels. Any other projection class must be explicitly authorized rather than inferred from array shape or a software default.

The coefficient ω_i comes from PTC. The approved PTC authority calls these scalar analysis and gridding coefficients and requires their family, units, normalization scope, lifecycle, and applied factors to be declared. They may be uniform, detector-based, sample-based, sensitivity-derived, or another named family. They are not inverse variances merely because their stored unit is inverse-square signal. A stronger precision interpretation requires separate calibration and covariance/independence evidence.

Here x_i is PTC's realized transformed signal. The paired PTC x/r components are upstream detector coordinates, not MAP response products; MAP carries only the facts needed by its interface and never reinterprets r as a response. An identity-like cleaning configuration still produces a realized PTC product. If that required product does not exist, MAP has no input.

The effective contribution is $G_{pi}\omega_i$. SCI-MAP first requires the PTC product to be available under PTC's exact declared identity and scope, then applies its own finite-signal, coefficient, projection, boundary, support, and use conditions. One layer cannot rescue a failure in the other. These conditions do not imply a universal hierarchy between occurrence-, detector-, observation-, and product-scoped failures. VAL can evaluate a named rule but does not invent either layer's policy. MAP then accumulates

$$N_p = \sum_i G_{pi}\omega_i x_i, \qquad Q_p = \sum_i G_{pi}\omega_i. \tag{4}$$

The numerator is not yet a sky estimate, and the denominator is a gridding normalization by default. Division occurs only where the adopted support policy permits a scientific map value.

Whether a fractional projection must conserve every interior sample through $\sum_p G_{pi} = 1$, how boundary loss is represented, and which non-one-hot classes are allowed are not fixed by the admitted upstream authority. They are therefore recorded as SCI-MAP-OD-008 rather than hidden behind the symbol G. Until resolved, every product must declare its actual projection and boundary convention, and no unrecorded conservation property may be assumed.

3 Projection, response, and correlated pixels

Fractional projection is useful because a sample can represent support that overlaps more than one pixel. The same operation also carries the sample's noise into each recipient pixel, correlating them.

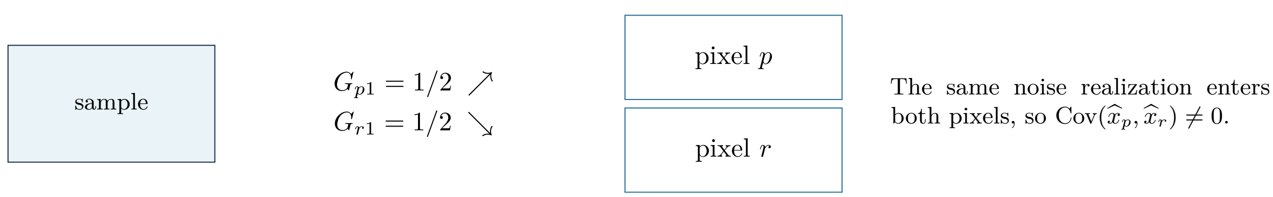

Figure 2: A fractionally projected sample creates shared noise support.

For a concrete calculation, take two independent unit-variance samples and, for this example, let $\omega_i = 1$ be their true inverse-variance coefficients. Split each sample equally into the same two pixels. In either pixel, $Q = 1/2 + 1/2 = 1$, but

$$\mathrm{Var}(\widehat{x}_p) = \frac{(1/2)^2 + (1/2)^2}{Q^2} = \frac{1}{2}, \qquad \phi_p = \mathrm{Var}(\widehat{x}_p)^{-1} = 2. \tag{5}$$

Thus normalization is 1 while formal marginal inverse variance is 2. In this deliberately symmetric example, the two pixel estimates are identical, their covariance is 1/2, and their correlation is unity: the second pixel does not constitute independent information. This is why projection, normalization, formal precision, and full covariance must be separate products.

4 Weights and uncertainty

For fixed upstream and map state, uncertainty propagates through the same map operator as the signal:

$$C_x = ANA^{\mathsf{T}}, \tag{6}$$

where N is the declared PTC-domain processed-sample noise covariance when it is available for the asserted domain. Equation (6) retains cross-pixel terms created by shared samples and any correlations already present in the timestream.

Four quantities answer different questions:

1. Q: the gridding denominator that normalizes the map;
2. a formal marginal inverse variance: valid only under its stated coefficient and covariance assumptions;
3. the full conditional map covariance or precision, including correlated pixels when available; and
4. empirical covariance, weights, or statistical significance calibrated by NOI from admitted realizations or data.

The raw map bundle reports what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, recoverable from lineage, summarized, incomplete, or unavailable. Missing correlations and nuisance terms are not zeros. Normalization, hits, exposure, and diagnostic weight are not covariance. Absence of a complete covariance model neither invalidates the map nor prohibits later analysis. A later covariance estimate is a new versioned product bound to the original map and processing identity.

The distinction is especially important when sample membership, weights, projection, response, or support is estimated from the same observation. The linear equations remain useful conditional descriptions, but they do not include selection uncertainty. A joint model, resampling study, or other separately justified treatment is needed for an unconditional statement.

Quantity	Owner/status	Science interpretation
Analysis coefficient ω	PTC, declared scalar family	Gridding/analysis coefficient; precision only if separately proved.
Normalization Q	MAP	Denominator of the ordinary estimator; not automatically uncertainty.
Conditional covariance	PTC input, MAP propagation	Fixed-state error covariance through projection and normalization.
Empirical weight/significance	NOI	Requires declared realizations, calibration, and validity evidence.
Calibration/response nuisance	SCI-CAL and response owner	Retained as quantified, lineage-resolvable, or unavailable; never silently zero.

Table 1: Ownership prevents a convenient numerical product from acquiring a stronger statistical meaning.

5 Coverage, support, and validity

The formal contract keeps eight facts separate. For science users they answer the following questions:

Science question	Product	Interpretation
Did projected samples geometrically reach this pixel?	Geometric hits	Geometry, not estimator admission.
How many valid weighted samples entered the estimator?	Accepted estimator hits	Exact contribution count.
How many complete observations entered this coadd pixel?	Contributing observation count	Observation bundles, not samples.
What detector-time was available before estimator-specific rejection?	Upstream-eligible exposure	Eligible detector-time accounting.
How much detector-time was retained by the estimator?	Retained exposure	Retained accounting, not precision.
Was the denominator large enough to normalize?	Numerical support	Lower support threshold.
Is the pixel inside the adopted science region?	Science-policy support	Stricter policy threshold.
Is the complete raw map value scientifically usable?	Final map validity	Required conjunction of identity, support, finite value, companions, and no product failure.

Hits are not validity, exposure is not precision, and positive normalization is not sufficient for final validity. A rectangular FITS image may contain storage values outside the scientific domain; the explicit support and validity products determine which pixels are measurements. Unsupported pixels are unavailable, not zero sky.

Provisional two-threshold policy. A lower threshold permits numerically safe normalization; a stricter threshold defines the adopted science region. Both are scaled from an order statistic of the finite positive normalizations in a declared map population. This is adopted policy, not a law of mapmaking or a proof of completeness, response, or noise quality.

Because the reference normalization is computed from a declared population, changing the map extent or population can change the science mask even when the local samples near a particular pixel are unchanged. That global behavior must be justified and validated for the intended use.

The formal equations compare Q_p with $Q_\star c/10$ and $Q_\star c$. The scientific owner has therefore fixed $c =$ `coverage_cut` as a dimensionless support-policy scalar. SCI-MAP-CI-001 records the resolved dimensional correction. This disposition does not determine the allowed numerical domain or the behavior for negative, zero, non-finite, or greater-than-one values; those questions remain open in SCI-MAP-OD-007.

6 Observation coaddition

V0.1 combines observation maps only when they already represent the same responded quantity on the same pixel grid. Each observation contributes as a complete bundle: signal, normalization, response state, covariance status, exposure, support, validity, WCS, and provenance are checked together.

The permitted placement is a centered integer embedding. A smaller map may sit inside a larger common grid only when each dimension differs by a nonnegative even number of pixels; half the difference is the integer offset. The shifted reference pixel must identify the same world coordinate. An odd shape difference is incompatible and the complete observation is rejected. There is no fractional shift, interpolation, reprojection, or implicit recentering in this ordinary branch.

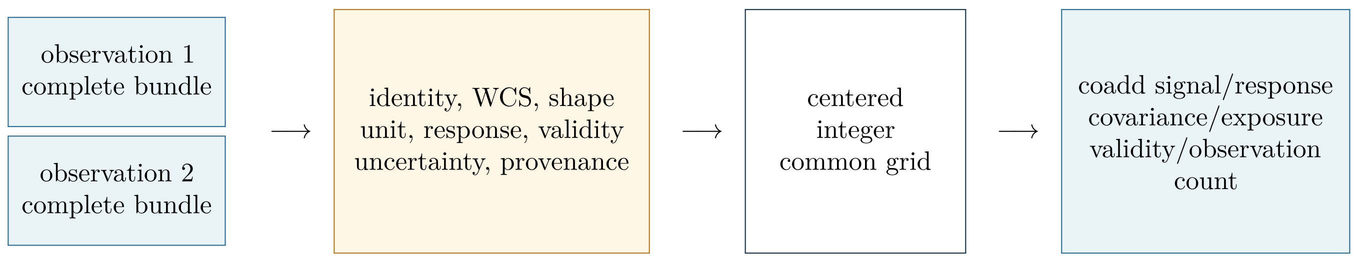

Figure 3: Complete-bundle admission and centered-integer placement. Any failed compatibility check rejects the observation bundle from the coadd before any of its valid pixels contribute.

This restriction is the declared scope of the ordinary v0.1 coadd, not a general scientific argument against mosaicking. Whether observation maps are cropped or padded upstream to a canonical grid, and which future package owns reprojection or mosaicking, are not fixed by the admitted authority and are recorded as SCI-MAP-OD-009. Until then, SCI-MAP performs neither operation.

The coadd propagates response and conditional covariance through the same observation membership and placement. Its denominator is again normalization, not automatically precision. Correlated generalized least squares, which may need negative coefficients and a full cross-observation covariance, belongs outside this positive-coefficient branch.

7 Units, WCS, products, and downstream use

SCI-MAP receives its scale and unit convention only through the realized PTC product; there is no direct SCI-CAL fallback. It preserves the coordinates, exact frame metadata, geometry, and coordinate validity supplied by ALIGN/AST and does not replace that authority with the shorthand phrase "J2000." PTC owns transformed-sample identity and availability, producer-local validity, and MAP-facing coefficient/covariance facts. MAP owns the target grid and its own projection, coefficient-admission, boundary, support, and use conditions. VAL may represent and evaluate named rules; it does not author those policies.

The accepted map-bundle authority is ADR 0009 and its 2026-08-05 owner amendment. The full binary64 typed or sidecar WCS is the lossless identity used for admission, common-grid comparison, and provenance. The WCS serialized in the FITS product is the representation ordinary scientific tools are expected to use for pixel-to-sky conversion. It must preserve axis sign, handedness, orientation, and the exact centered-integer reference-pixel relations, and must agree with the lossless authority within 0.1 arcsec maximum sky separation. The bound is cited from that owner-approved representation authority; this rationale does not invent a physical justification for it.

The FITS WCS is sufficient for ordinary pixel-to-sky use at its declared serialization fidelity, while the sidecar retains the exact WCS identity used for admission, conformance, provenance, and coadd comparison. A tolerance never authorizes a different sky grid, fractional shift, interpolation, or reprojection.

Science-user state	Interpretation and required action
Valid map; response available	Use only for the response class and processing lineage actually declared.
Valid numerical map; response unavailable	Preserve the honest limitation. The original bundle makes no unsupported response-dependent claim; later analysis may attach a new versioned response or corrected product.
Formal weight; empirical significance unavailable	Use only according to its declared role—normalization or formal precision—and do not call it significance.
Covariance summarized or unavailable	State meaning, domain, and limits. The map remains valid; later covariance work is a new bound product.
Unsupported pixel	Unavailable, not zero sky.
Raw-invalid pixel	No downstream finite value may promote it to raw-valid.
Coadd-incompatible observation	Entire observation bundle is excluded; no partial coadd.
WCS or unit incompatibility	Reject combination rather than silently convert, recenter, or relabel.

Downstream packages inherit the raw unit, response, covariance, validity, WCS, and provenance. NOI owns empirical noise and significance; FLT owns filtered response, covariance, and validity while preserving the immutable raw parent; source, Pointing/OOF, and Beammap packages own their scientific inference.

8 Validation

Validation must keep algebra, software conformance, representation fidelity, observational performance, and production readiness separate. The required science program is:

1. analytic estimator tests for constants, unequal coefficients, and exact hand calculations;
2. one-hot and fractional projection tests for response, formal variance, and cross-pixel covariance;
3. support, boundary, global-population, and unsupported-pixel tests;
4. complete-bundle compatibility and centered-integer coadd placement tests;
5. sequential/parallel conformance: floating results agree within a preregistered bound appropriate to operation count and conditioning, while identities, counts, masks, support, offsets, and product cardinality agree exactly;
6. synthetic point, extended, gradient, edge, and variable-coverage response tests; and
7. observational amplitude, shape, position, covariance, and repeatability checks against independent references.

No test, implementation inspection, reduction, or production qualification was executed for this authorship revision.

9 Open scientific decisions

The compact register in Appendix B preserves OD-001 through OD-009. SCI-MAP-CI-001 is resolved: `coverage_cut` is dimensionless. OD-007 remains open only for its numerical domain and failure behavior. The other open items expose genuine upstream gaps rather than asking implementation questions.

Rationale r0.4 integrates the owner-authorized PTC-to-MAP handoff correction. Further revisions should follow an owner decision, a normative contract change, new validation evidence, or a newly identified scientific inconsistency–not stylistic churn.

References

[1] *SCI-MAP v0.1 Formal Scientific/Engineering Contract*, canonical shared authority and amended r0.4 formal view.
[2] *SCI-CAL v0.1 Science-Team Rationale*, revision r0.3.
[3] *SCI-PTC-001 D004 Owner Amendment*, approved 2026-08-09, for scalar coefficient and covariance semantics.
[4] *Citlali Scientific Conventions*, map-bundle and ownership authority.
[5] ADR 0009, *Science-Map Bundle Admission and Validity*, including the 2026-08-05 WCS/FITS persistence amendment.

A Compact rationale-to-contract crosswalk

The separate `CROSSWALK.md` carries the detailed routing. The formal contract remains normative; this appendix does not replace or paraphrase any requirement or prediction as authority.

Formal authority	Science-team location	Continuing formal home
REQ-001–010	Executive summary; Sections 1–2	Version, input, identity, projection, coefficient, lifecycle, admission.
REQ-011–018	Sections 2–3	Accumulators, valid-domain publication, response, absolute mode.
REQ-019–024	Section 4	Conditional covariance, formal weight, nuisance state, statistical labels.
REQ-025–035	Section 5	Eight facts, exact threshold selector, support, raw validity.
REQ-036–042	Section 6	Complete bundles, centered-integer coadd, response/covariance, GLS boundary.
REQ-043–048	Section 7	Product identity, WCS, provenance, failure, publication.
REQ-049–052	Sections 7–8	Raw parent, consumer boundary, fixed realization operator, claim layers.
PRED-001–025	Section 8	Full formal prediction inventory and exact evidence classes.
OD-001–009	Section 9; Appendix B	External scientific-owner decision ledger.
SCI-MAP-CI-001	Section 5	Resolved dimensionless classification and bounded normative amendment.

B Compact scientific-owner decision register

All items are open. The exact questions, interim behavior, authority needed, and affected clauses live in `SCIENTIFIC_OWNER_DECISION_LEDGER.md`.

ID	Issue	Conservative status while open
OD-001	Threshold rationale	Adopted policy only; no physical optimality or completeness claim.
OD-002	Threshold change authority	Formula and population are frozen until an owner-approved successor.
OD-003	Response obligations	Original claims use only response information actually provided; later response work is a bound versioned product.
OD-004	Covariance persistence	Representation, meaning, domain, and omissions are explicit; absence does not invalidate the map.
OD-005	Observation-map publication	Follow the effective required-output policy; infer no universal storage mandate.
OD-006	Pointing/OOF reuse	No mode-specific authority follows from ordinary arithmetic alone.
OD-007	`coverage_cut` numeric domain	Dimensionless status is fixed; no numerical domain or boundary behavior is inferred.
OD-008	Projection normalization and boundaries	Declare the actual class and boundary convention; assume no unapproved conservation property.
OD-009	Common-grid preparation and future mosaicking	Odd-difference inputs reject; MAP performs no crop, pad, reprojection, or mosaic absent new authority.

C Document authority

Rationale revision r0.4 integrates the PTC-to-MAP handoff correction and retains the scientific-owner disposition of SCI-MAP-CI-001. The scientific contract remains v0.1, with 52 canonical requirement IDs and 25 canonical prediction IDs unchanged. Their bounded amended clauses now classify `coverage_cut` as dimensionless while leaving its numerical domain and failure behavior open under OD-007. The formal contract and engineering conformance specification remain the normative authorities for audit, repair, re-audit, and exact product behavior.